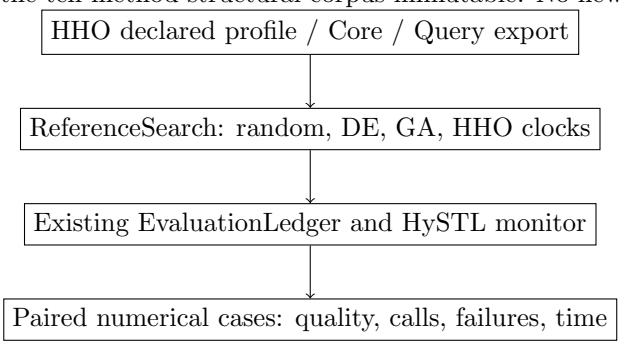
HHO numerical pilot, pre-run protocol

Created-at: 2026-10-08T18:48+02:00 (Europe/Budapest). ETA: 60–90 minutes for profile, hand-checked branch tests, numerical pilot and write-up. Expected run under 10 minutes; hard campaign command wall budget 600 s.

Scope First executable method beyond the frozen DE/GA/random references. A bounded, source-derived HHO variant, not an unchanged MATLAB implementation. Keep the ten-method structural corpus immutable. No new parser, canonical graph, optimization framework or monitor.



HHO choices Sequential in-place proposals and population means; leader updated only by population scans. Every objective call, including repeated parent comparisons, is charged. Repair every evaluator input by componentwise clamp; store the repaired accepted trial. Pending unconditional proposals are repaired at the next scan. Python binary64/Random, Mantegna beta=1.5 without extra scale. Zero normal divisor/nonfinite proposals fail explicitly. Return host best-ever, with last-scored slot snapshots separately retained. Stop before B+1, including mid-comparison; no unscored score fabrication.

Clock control Single implementation, two named settings. Iteration clock uses T=ceil(B/N), factor=2*(1-t/T). Evaluation clock uses factor=2*(1-(count-N)/(B-N)) at the end of each population scan and freezes it for that update sweep. No update at B=N. Source RNG order and conditional branches are documented; no MATLAB bitwise-RNG equivalence claim.

Matrix frozen before measurement Sphere, Rosenbrock, Rastrigin; D=2,5; origin, shift, shift+pairwise orthogonal rotation; 30 seeds 0..29; random, DE, GA, HHO-evaluation, HHO-iteration. N=8, B=80, bounds [-5,5], F=.5, CR=.9. Total 2700 cells, 216000 search calls plus 2700 separately charged independent validation calls if all cells complete. Same initial population per paired block. Fixed shift from local problem RNG seed 20261008+D; paired rotations by pi/5. No tuning or selection from pilot outcomes. These are custom analytic cases, not BBOB/CEC.

Metrics and resources Per-block median/IQR best-so-far curves, success at error below 1e-6, paired wins/ties/losses with relative tolerance 1e-12; failures remain in denominator. Record objective/total instrumented search time separately. No cross-family universal ranking. Serial host worker on Amaterasu; expected peak under 1 GiB; no inference/GPU/physics. JSON/CSV, PNG/SVG and a 2D actual-evaluation GIF. No objective calls solely for background plotting. Monitor uses actual elapsed traces and is separately timed.

Files Transformation: src/metaheuristic/profiles.rs, new profiles/hho.rs if table separation is needed, mod.rs exports, bin/metaheuristic_contract.rs; data/metaheuristics/hho_sequential_budget.hymeko; tests/hho_contract.rs; README and SYMBOLS. Hystl research/metaheuristics-pilot: hho_reference.py, test_hho_reference.py, numerical_campaign.py, test_numerical_campaign.py; reference_search.py, bind_operator_ids.py, README.md, PROTOCOL.md, HHO.md. Plans and report artifacts.

Checks Static contract mutations/roundtrip; frozen DE/GA export parity; independent small arithmetic cases for every HHO branch, conditional Levy and repeated calls; B=N and mid-comparison stopping; initial ledger/config admission; deterministic semantic trace replay; metamorphic objective transform tests. Pilot preflight at complete B80 precedes the matrix. Check source/binary/data hashes, Clippy, Python lint/type checks and clone deltas. No new dependencies, versions, worktrees or protected items.

Risk and rollback NP8/B80 is deliberately a small pilot, not a scientific superiority result. Coordinate-origin and energy-schedule sensitivity are primary controls. Trace instrumentation affects timing. Preserve original APIs/caps and historical run artifacts; revert only task hunks from pre-edit snapshots. No commit/push.